

Helvågslikriktning och Spänningsaggregat

Viktor Öwall



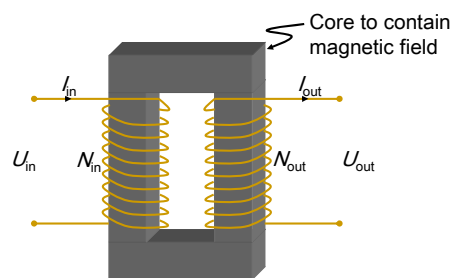
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Elektromagneter en ström i en ledare skapar ett magnetfält



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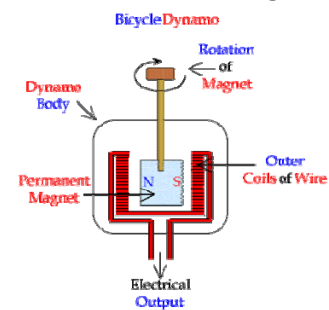
Transformatorn (AC till AC)



En ström skapar ett magnetfält.
En ledare i ett varierande magnetfält kommer
att få en inducerad ström/spänning.

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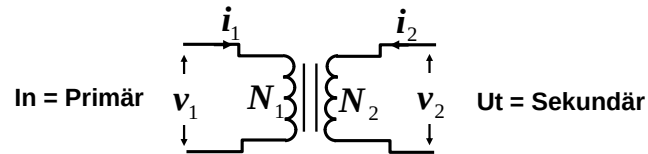
Dynamon



En ledare som rör sig i ett varierande magnetfält kommer att få en inducerad ström.

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Transformatorn



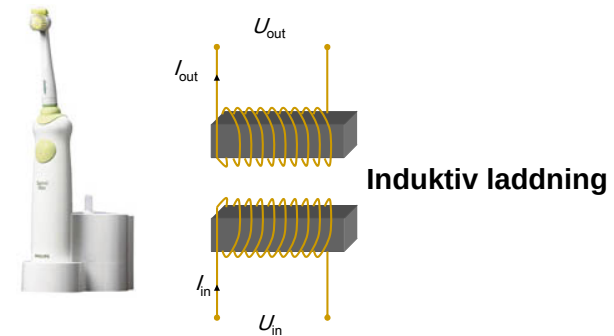
$$\frac{v_1}{v_2} = \frac{N_1}{N_2}$$

$$i_1 N_1 + i_2 N_2 = 0$$

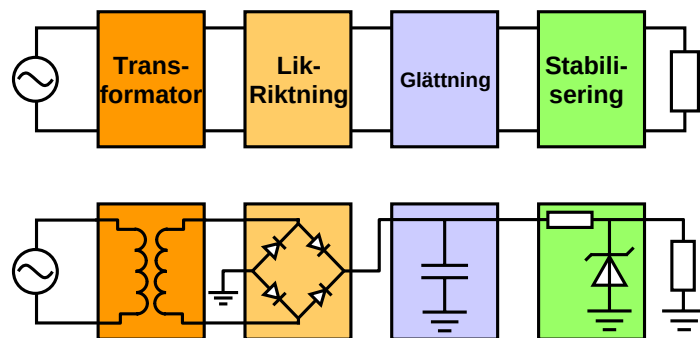
Används för att förändra spännings- och/eller strömförhållande, t.ex. för att sänka nätspänningen till en lämplig spänning för integrerade kretsar, stereo, ...

Även skyddstransformatörer.

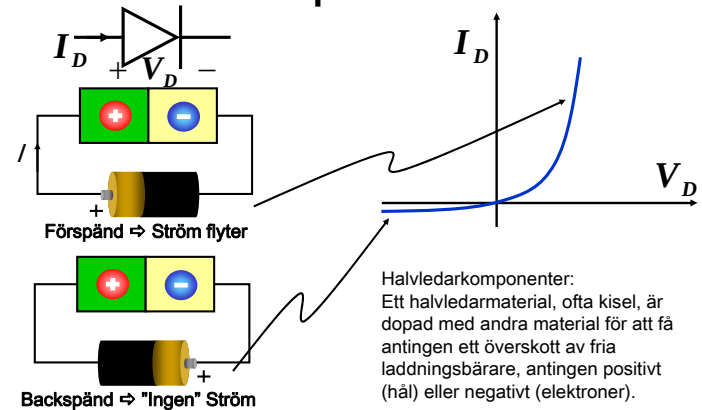
Ladda Tandborsten



Spänningsaggregat

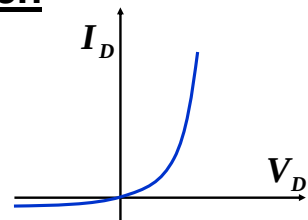
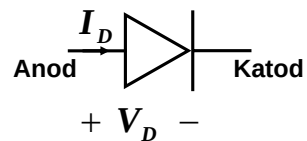


Halvledarkomponent: Dioden



Dioden

Halvledarkomponent



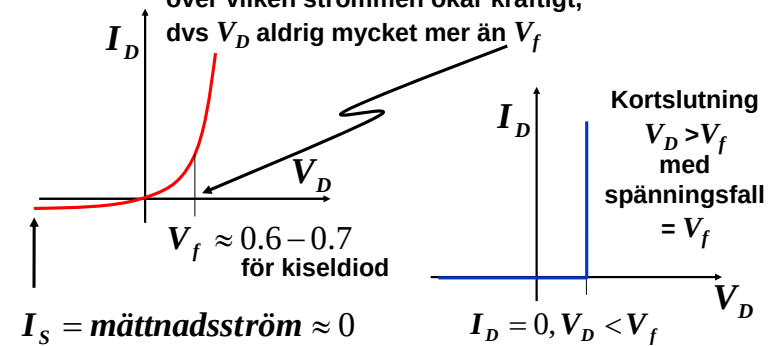
$$I_D = I_S (e^{V_D/V_T} - 1)$$

$$V_T = \text{termiskspänning} = \frac{kT}{q}$$

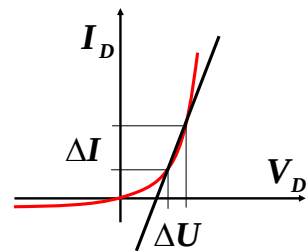
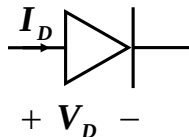
$$I_S = \text{mättnadsström} \approx 10^{-15} \text{ A}$$

Dioden, förenklad modell

Tröskel- eller Knäspänning, V_f ,
över vilken strömmen ökar kraftigt,
dvs V_D aldrig mycket mer än V_f

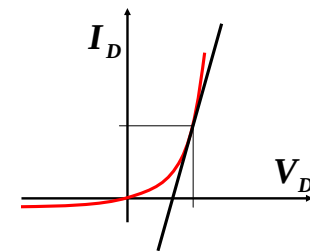
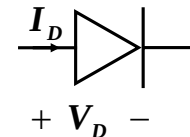


Dynamisk resistans

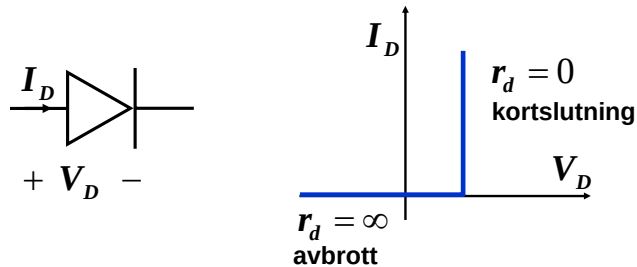


$$r_d = \frac{\Delta U}{\Delta I}$$

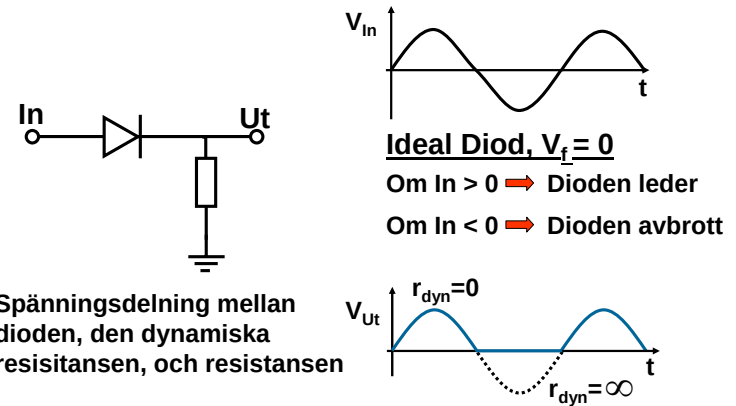
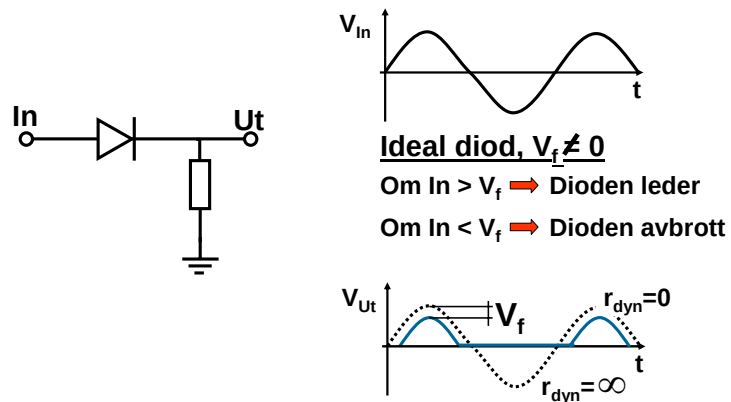
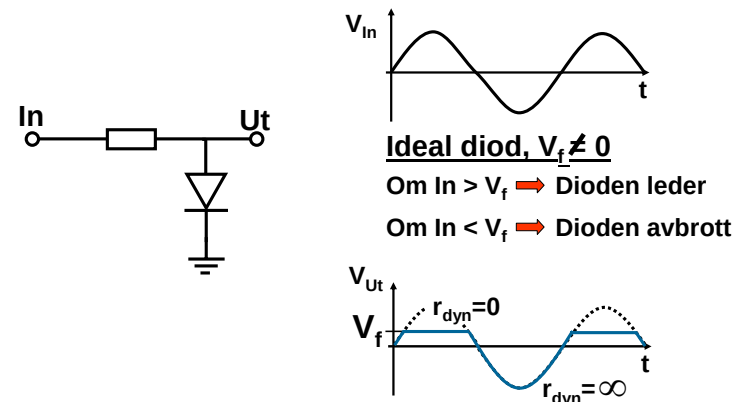
Dynamisk resistans



Tangenten = Derivatan då $\Delta \Rightarrow 0$

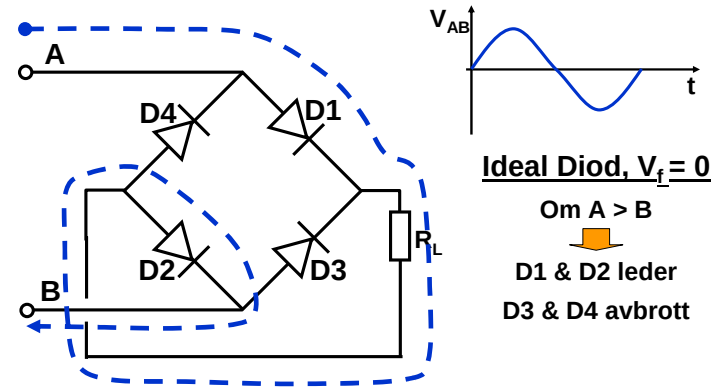
Dynamisk resistans

$$r_d = \frac{\Delta U}{\Delta I} \quad \begin{cases} \text{framriktning} & r_d = 0 \\ \text{backriktning} & r_d = \infty \end{cases}$$

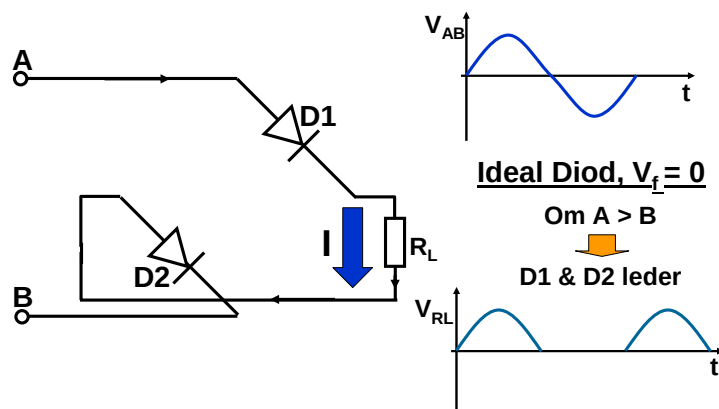
HalvvågsikriktningHalvvågsikriktning ($V_f \neq 0$)Halvvågsikriktning ($V_f \neq 0$)

Med halvågslikriktning
"slänger vi bort"
halva signalen!

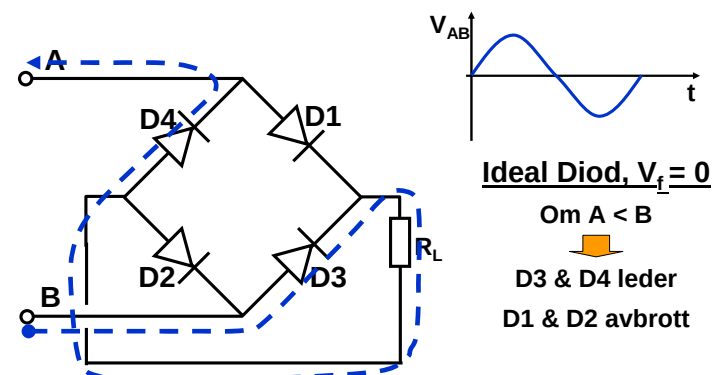
Helvågslikriktning

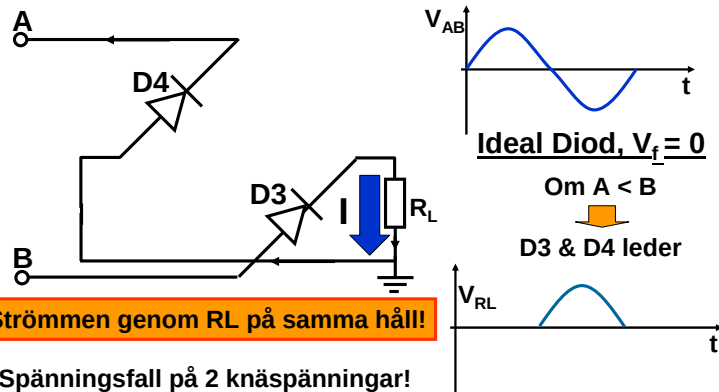


Helvågslikriktning, en halv period

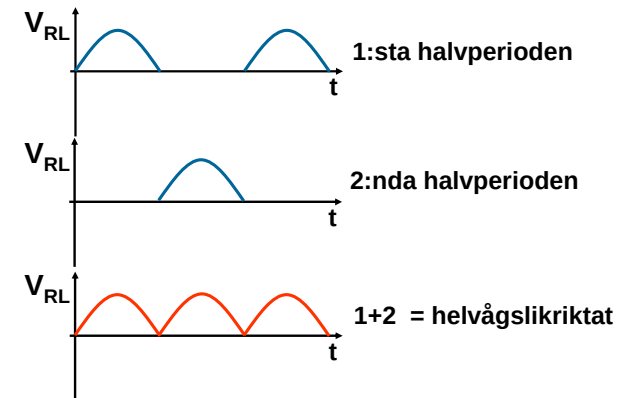


Helvågslikriktning, andra halvperioden

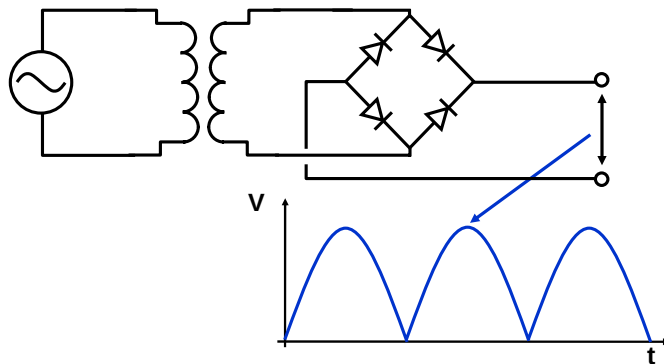


Helvågsl riktnig, andra halvperioden

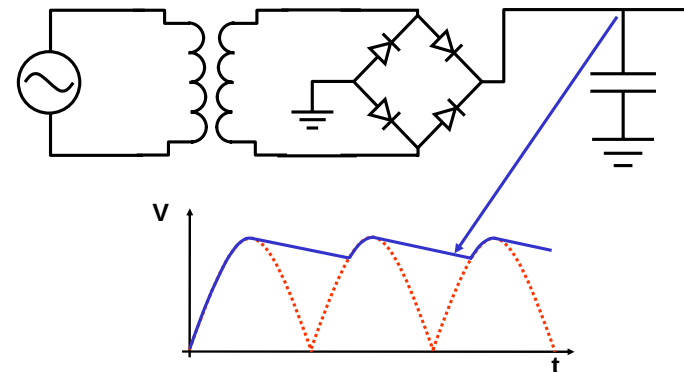
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Helvågsl riktnig, totalt

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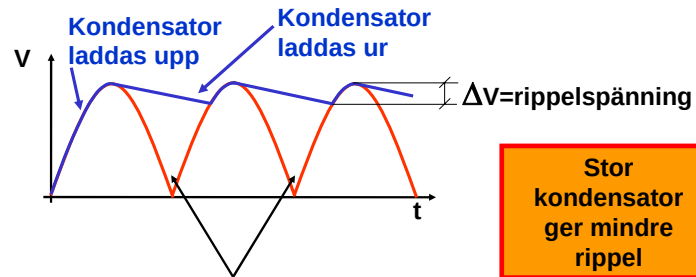
Glättning

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Glättning

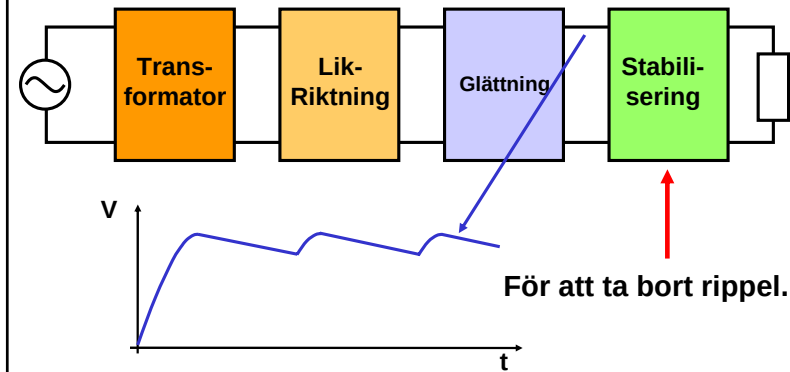
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Glättning

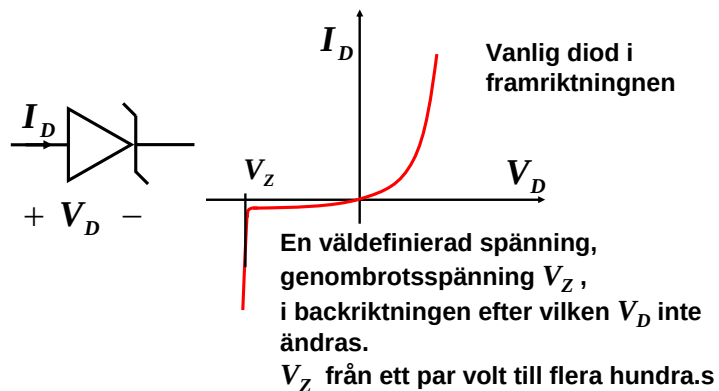


Ta bort "dalarna" för att få en likspänning
 "Lagra" värdet på kondensator
 = Glättning

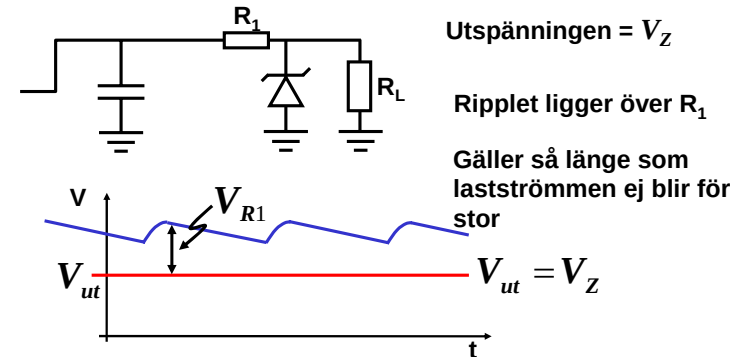
Stabilisering



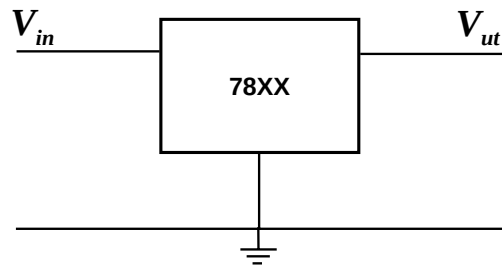
Zenerdioden



Zenerstabilisering



Regulatorstabilisering



Fast utspänning: XX = 05 $\Rightarrow$ 5V
 XX = 15 $\Rightarrow$ 15V

Regulatorstabilisering

